'Open Source' Technology Readiness
The case of Distributed Recycling

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Agenda I

Technology Readiness: A programmatic figure of merit (FOM)

- Technology Readiness Levels (TRL- $1-9$)
- FOM $\rightarrow$ *effective assessment* and *communication* of the maturity of new technologies.
- Adoption by International Commissions
- Tool $\rightarrow$ *funding agencies*
- Traditional intellectual property (IP) mechanisms are in place

However, what about in 'collaborative projects'?

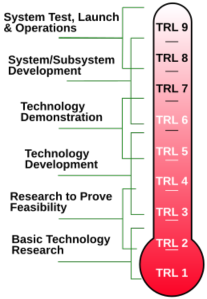


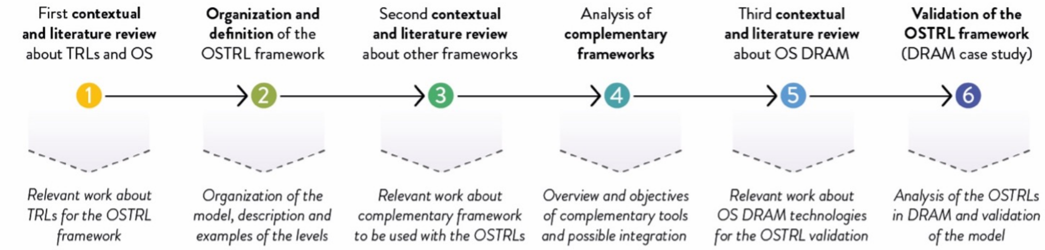
Figure 1: TRLs developed by the National Aeronautics and Space Administration (NASA) (Mankins, 2009)

Problem: Co-creation / Collaborative / Open Source

However, traditional TRL reflects conventional product development processes and traditional IP mechanisms.

- Challenges in the 1) system complexity, 2) planning and review, and validity of assessment 3) (Olechowski et al., 2020).
- They are not fully adequate to assess the Readiness of technologies developed in open and collaborative settings (Yfanti and Sakkas, 2024)
- There is a need of drivers (external and internal) to help companies to implement open and environmentally sustainable innovations (OESI) (Janik and Ryszko, 2025)

Methodology



Open Source TRL (OSTRL): Theory

It introduces two new elements [...*regarding traditional TRL*] as key factors to assess the readiness of a technology :

IP openness
: which refers to the legal freedom to *study, replicate, distribute, and modify the technology* without imposing restrictive IP constraints, e.g., patent exclusivity or license-based limitations.

Documentation openness
: which refers to the *availability, completeness, and accessibility of technical documentation* required to study, replicate, distribute, and modify the technology, e.g., source design files, bill of materials (BOM), or assembly instructions.

Open Source TRL (OSTRL): Principles

Domains of readiness for open design:

- *Product openness*, $\rightarrow$ the design outcome,
  - e.g., designs or off-the-shelf components (OSTRLs 1-7).
- *Process openness*, $\rightarrow$ to the design process,
  - e.g., manufacturing processes or software (OSTRLs 7-9).

Readiness functions:

1. ***Transparency*** $\rightarrow$ the freedom to study and understand how the technology works.
2. ***Replicability*** which $\rightarrow$ the freedom to make and replicate the intended technology, retaining the intended functions in the replicas.
3. ***Reusability***, $\rightarrow$ the freedom to distribute and use the technology without any functional limitation.
4. ***Accessibility***, $\rightarrow$ the freedom to modify and improve the technology without any restrictions.

Open Source TRL (OSTRL): Proposition

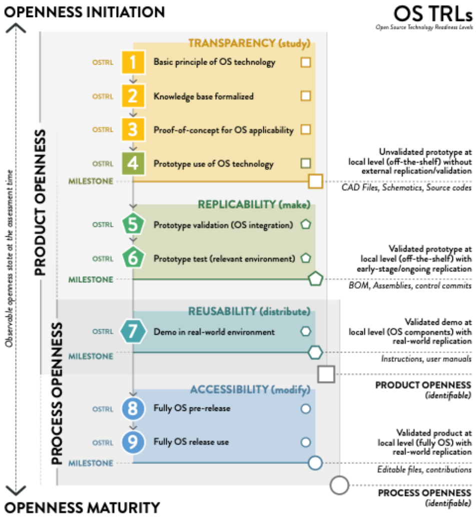


Figure 2: Methodological proposition

Table 2. OSTRL framework: domains of readiness, readiness functions, OSTRL levels, expected milestones, and minimum documentation when achieving the milestone (last level of each OSTRL function).

Domains of Readiness [43] and OSTRL functions [42]	OSTRL Levels [5] and Definition	Explanation	Milestone (OS readiness) and minimum expected OS documentation
Product openness. Transparency (freedom to study)	**OSTRL 1.** Basic principles observed and reported in scientific or proprietary literature.	The lowest level of OS technology readiness. Fundamental research begins to select the technology to be translated into OS.	**End of OSTRL 4.** Unvalidated prototype at local level (off-the-shelf) without a third-party external replication (internal validation). **Expected:** CAD models and schematics (OS hardware); Source code (commented), architecture documentation (OS software), explanation of logic flow.
	OSTRL 2. Technology concept and/or application is formulated in the OS context.	Practical OS applications and concepts can be designed, starting with R&D. Applications are speculative and may be unproven.	
	OSTRL 3. Analytical and experimental critical function and/or characteristic proof of concept using off-the-shelf* or code/scripts.	Active R&D is initiated, including laboratory studies to validate predictions regarding the technology built using off-the-shelf* or software code/scripts.	
	OSTRL 4. Component validation in a laboratory environment using off-the-shelf* or code/scripts.	Basic technological components are integrated to establish that they will work together using off-the-shelf* or software code/scripts.	
Product openness. Replicability (freedom to make)	**OSTRL 5.** Component validation in relevant environments using off-the-shelf* or code/scripts.	The basic components are integrated with reasonably realistic supporting elements to be tested in a simulated environment using off-the-shelf* or software code/scripts.	**End of OSTRL 6.** Validated prototype at local level (off-the-shelf) with at least one third-party early-stage/ongoing replication. **Expected:** CAD models, schematics, BOM, and assemblies (OS hardware); build instructions, dependency list, environment configuration (as a text file), version control commits (OS software).
	OSTRL 6. System model or prototype demonstration in a relevant environment using off-the-shelf* or code/scripts.	A representative model or prototype system is tested in a relevant environment using off-the-shelf* or software code/scripts.	
Product and process openness. Reusability (freedom to distribute)	**OSTRL 7.** System prototype manufactured potentially using only OS components or OS software code/scripts.	A prototype OS system that is near, or at, the planned operational system using only OS components or OS software code/scripts.	**End of OSTRL 7.** Validated demo product at local level (OS components) with at least one third-party real-world replication. **Expected:** CAD models, schematics, BOM, assemblies, and instructions (OS hardware); Software license file, APIs, user manual interoperability documentation, (OS software).
Process openness. Accessibility (freedom to modify)	**OSTRL 8.** System completed using only OS components, processes, or software code/scripts tested and demonstrated.	The system has been proven to work in its final form and, under expected conditions, is fabricated only with OS processes and components or OS software code/scripts.	**End of OSTRL 9.** Validated product at local level (fully OS) with real-world replication and modifications. **Expected:** CAD models, schematics, BOM, assemblies, instructions, and editable files (OS hardware); access policy, contribution guidelines, issue tracking repository, pull requests for new versions (OS software).
	OSTRL 9. System proven to use digital replication in a pure OS environment successfully operated, and modifiable.	The final version of the system has been used under actual deployment conditions as a complete OS digital replication on only OS tooling or OS software, allowing for modifications.	

* = Potentially proprietary components or processes for fabrication.

Figure 3: Detail structure

Distributed Recycling via Additive Manufacturing (DRAM)

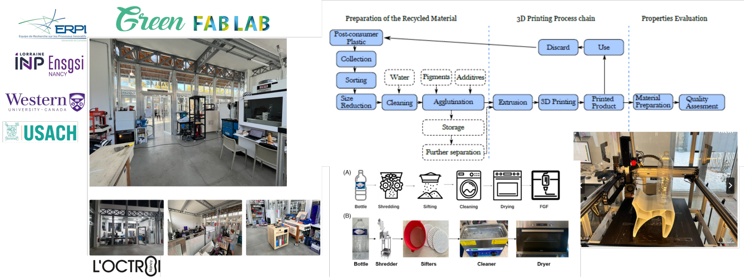
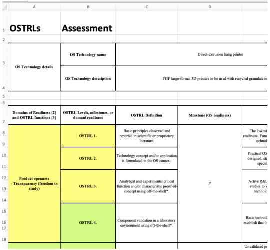


Figure 4: Distributed Recycling via Additive Manufacturing

- Gonzalez, C.S. et al (2025). *Strategies for recycling multi-material polymer blends for additive manufacturing*. Sustainable Materials and Technologies 44, e01430. https://doi.org/10.1016/j.susmat.2025.e01430
- Caceres-Mendoza, C., et al. (2023). *Life cycle assessment of filament production in distributed plastic recycling via additive manufacturing*. Cleaner Waste Systems 5, 100100. https://doi.org/10.1016/j.clwas.2023.100100
- Cruz Sanchez, F.A., et al. (2020). *Plastic recycling in additive manufacturing: A systematic literature review and opportunities for the circular economy*. Journal of Cleaner Production 264, 121602. https://doi.org/10.1016/j.jclepro.2020.121602

OSTRL of DRAM



OSTRLs	Assessment			
OS Technology details	OS Technology name	Direct-extrusion hang printer		
	OS Technology description	FGF large-format 3D printers to be used with recycled granulate m		
Domains of Readiness [2] and OSTRL functions [3]	OSTRL Levels, milestones, or domani readiness	OSTRL Definition	Milestone (OS readiness)	
Product openness - Transparency (freedom to study)	OSTRL 1.	Basic principles observed and reported in scientific or proprietary literature.	//	The lowest readiness. Func technol
	OSTRL 2.	Technology concept and/or application is formulated in the OS context.		Practical OS designed, sta specul
	OSTRL 3.	Analytical and experimental critical function and/or characteristic proof-of-concept using off-the-shelf*.		Active R&D studies to v technolo
	OSTRL 4.	Component validation in a laboratory environment using off-the-shelf*.		Basic technol establish that th
				Unvalidated p

Figure 5: Individual evaluation of technologies

Templates at OSF Repository
https://osf.io/huwbr/overview

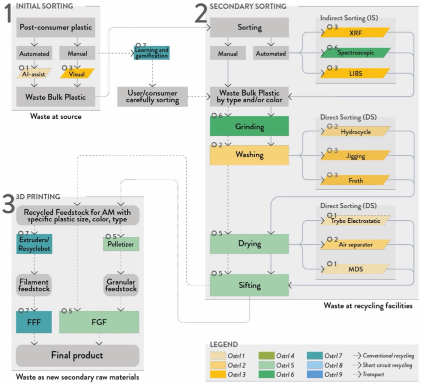


Figure 6: Open Source Technology Readiness Level Assessment of DRAM

Conclusions & Future

Romani A. et al (2026), *Open Source Technology Readiness Level Assessment and Application to the Case Study of Distributed Recycling for Additive Manufacturing* Technological Forecasting and Social Change, (Submitted)

Perspectives:

- Other Open Source Technologies / Ecosystems?
- Integration of data-driven / AI approaches as ex-ante evaluation of OSTRL?
- Industry 5.0 / Right to Repair $\longleftrightarrow$ OSTRL

Questions

Merci beaucoup de votre attention. Bonne année 2026 et Bonne Galette des Rois !

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Mankins, J.C., 2009. Technology readiness assessments: A retrospective. Acta Astronautica 65, 1216–1223. https://doi.org/10.1016/j.actaastro.2009.03.058

Olechowski, A.L., Eppinger, S.D., Joglekar, N., Tomaschek, K., 2020. Technology readiness levels: Shortcomings and improvement opportunities. Systems Engineering 23, 395–408. https://doi.org/10.1002/sys.21533

Yfanti, S., Sakkas, N., 2024. Technology Readiness Levels (TRLs) in the Era of Co-Creation. Applied System Innovation 7, 32. https://doi.org/10.3390/asi7020032